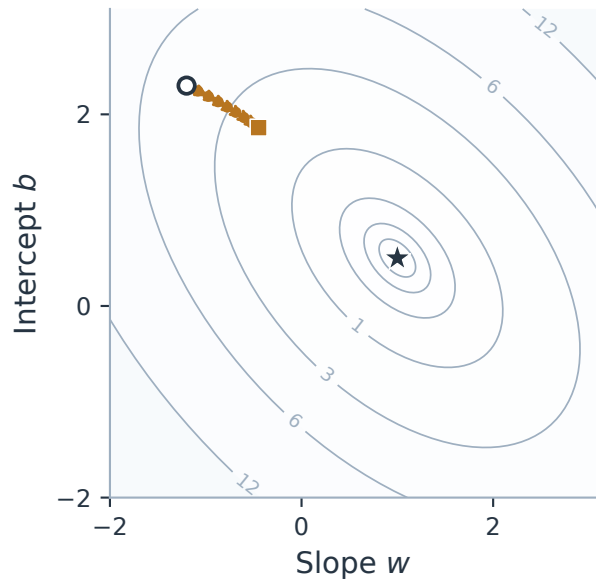


Same loss surface, different learning rates

$$\mathcal{L}(w, b) = \frac{1}{n} \sum_i (wx_i + b - y_i)^2 \quad | \quad \text{Same start and 8 updates in every panel}$$

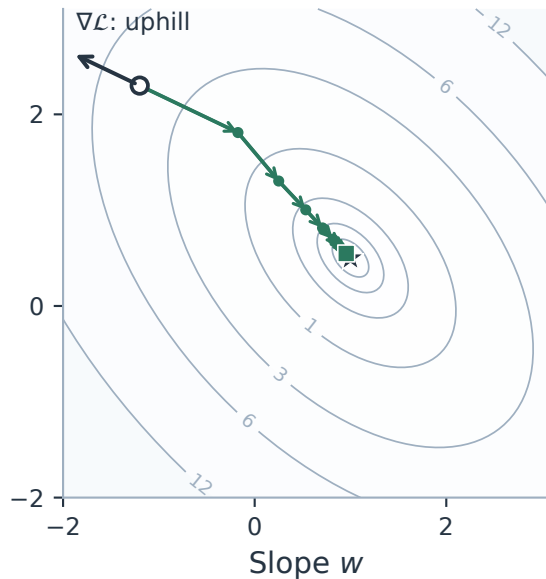
Small: $\eta = 0.04$



Slow progress

MSE: 4.48 $\rightarrow$ 2.136

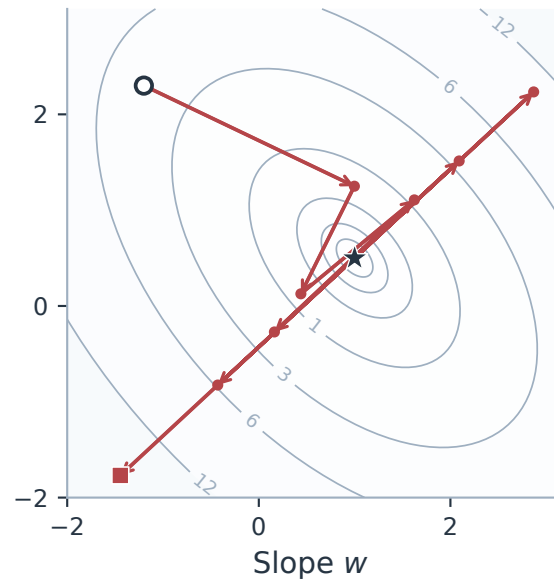
Sensible: $\eta = 0.35$



Approaches the minimum

MSE: 4.48 $\rightarrow$ 0.002

Excessive: $\eta = 0.75$



Overshoots, then diverges

MSE: 4.48 $\rightarrow$ 17.123

○ Common start ★ Minimum ■ After 8 updates

Contour labels: MSE